AIMAAN KHAN

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SUMMARY

ML engineer and M.S. Applied Data Intelligence student at SJSU. I take models from notebook to something people use - reusable pipelines, CLIs, web apps and tests - with industry ML experience on cybersecurity threat data and two published papers.

EDUCATION

San José State University **Aug 2026 - Present**
Master of Science, Applied Data Intelligence *San Jose, CA*
Coursework: Python Programming & Data Structures (DATA 200), Database Technologies (DATA 201)

University of Mumbai **2021 - 2025**
B.E. Computer Engineering, Honors in Cyber Security | *GPA 3.37 / 4.0 (WES; 7.79 / 10)* *Mumbai, India*

SKILLS

ML: scikit-learn, NLP (spaCy NER, TF-IDF), PyTorch (basics), model evaluation, feature engineering
Deployment: Saved pipelines (joblib), Streamlit, Flask, FastAPI, REST APIs
Engineering: Python, SQL, JavaScript, Git, pytest, Linux
Data & Cloud: pandas, NumPy, MongoDB, AWS | **Spoken:** English, Hindi

EXPERIENCE

Software Developer Intern - cloudyBoss Pty Ltd **Aug 2023 - Jun 2024**
- Trained endpoint detection and response (EDR) models on STIX-compliant threat signals to automate threat detection.
- Contributed ML components to the ANIMA AI security engine; earned cBX84 accreditation in machine learning and solution design.
- Designed a universal data denormalization module in SQL that turns federated relational data into ML-ready tables.

Python Developer Intern - BeamOptics Scientific Pvt. Ltd. **Jul 2025 - Oct 2025**
- Developed a statistical model on real-time light-sensor data that estimates the MBRT result before the test completes.
- Built the complete software stack for an automated Methylene Blue Reduction Test (MBRT) milk-quality machine on Raspberry Pi 4 - sensor integration, backend logic and a Kivy touch UI.
- Owned lab testing, debugging and deployment through to a ready-to-deploy machine.

Web Developer Intern - Prism Cybersoft Pvt. Ltd. **Jun 2024 - Jul 2024**
- Built a responsive documentation website for SOHAM software using JavaScript and the MERN stack.

PROJECTS

Cyber-Attack Detection | *Python, scikit-learn, pandas, Streamlit, pytest*
- Built Random Forest models on 82K UNSW-NB15 network flows: 97.6% accuracy and 0.997 ROC-AUC for attack detection, plus 10-class attack categorization.
- Found and removed label leakage that inflated the original model to a false 100%; shipped saved pipelines, a CLI, a Streamlit app and a pytest suite.

Toxic Comment Detector | *Python, scikit-learn, NLP, Flask*
- Built a multi-label classifier for 6 toxicity types on 159K comments (TF-IDF word + character n-grams, Logistic Regression): 0.978 mean ROC-AUC on 64K unseen test comments.
- Served it through a Flask web app and JSON API with request limits and XSS-safe rendering.

AI Timetable Scheduler | *Node.js, Express, MongoDB, FastAPI, spaCy*
- Trained a custom spaCy NER model so staff can edit timetables in plain English; published in JETIR as first author.
- Built a constraint-based scheduler that generates clash-free timetables for a whole department, with global teacher and room conflict checks and Excel export.

PUBLICATIONS

- A. Khan et al., "AI-Powered Timetable Scheduler and Management," JETIR, vol. 11, no. 10, Oct. 2024.
- M. Gupta, M. Siddique, A. Khan, P. Thorat, "IoT-Based Vehicle Location and Speed Monitoring for Parental Peace of Mind," JETIR, vol. 11, no. 4, Apr. 2024.

LEADERSHIP & VOLUNTEERING

Team Lead - two academic ML/software projects 2023 - 2025
Volunteer - DevFest Bay Area 2026 (Google Developer Groups) Oct 2026

CERTIFICATIONS

AWS Educate Machine Learning Foundations · Google AI Essentials · IIT Bombay AI & ML Workshop · AWS Academy Cloud Foundations